

Amir Khajavirad

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PROFESSIONAL SUMMARY

Biomedical engineer with hands-on experience across robotics and embedded systems development, biomedical device testing, and clinical data analysis. Builds and maintains Raspberry Pi based robotic platforms integrating sensors, motors, GUI controls, networked multi-device communication, and a secure over-the-air update pipeline for laboratory research, using AI-assisted development tools in daily implementation and debugging. Combines mechanical design and biomechanical testing experience with applied statistics and machine learning for clinical and experimental datasets. Comfortable working across hardware, software, and data layers to deliver reliable, reproducible research tools.

TECHNICAL SKILLS

Programming and Systems: Python (standard library, object-oriented design), Java (Android development), C++ (foundational, embedded/Arduino), SQL, Bash, Linux, Multithreading, State Machines, Atomic File Operations, Subprocess Management, File-Based Inter-Process Communication (IPC), JSON and CSV Data Handling

AI-Assisted Development: GitHub Copilot, Claude, ChatGPT: Code Generation, Debugging, Refactoring, Documentation

Embedded and Robotics Systems: Raspberry Pi 4/5, GPIO, SPI, UART/Serial Communication, RFID Integration, HX711 Load Cell Integration, Photo Interrupter Sensors, Stepper Motor Control, Sensor Integration, Hardware-Software Integration

Robotics Software and Control: Sensor-Driven Control Logic, State Management, Hardware Calibration, Motor Homing and Positioning, Event and Outcome Tracking, Safety Interlocks, Experimental Automation

GUI and Interfaces: Python Tkinter, Touchscreen GUI Development, Touch-Optimized Components (Scrollable Pickers, On-Screen Keyboards), Guided Operator Workflows, Real-Time System Monitoring

Networking and Connectivity: TCP Socket Communication (Pi-to-Pi Protocol), REST APIs, NetworkManager, nmcli, journalctl, 802.1X and WPA2-Enterprise Wi-Fi (eduroam), Wi-Fi Configuration

Software Reliability, Deployment and Security: Git, GitHub, CI/CD (GitHub Actions), Over-the-Air (OTA) Update Pipelines, SHA256 Integrity Verification, Semantic Versioning, Versioned Release Packaging, Preflight Validation, Staging, Backup and Rollback, Unit Testing with Mocks, Profiling (cProfile), Secure Credential Handling, Runtime Logging

Data Acquisition and Analysis: RFID and Sensor Data Acquisition, Experimental Data Management, Time Series Logging, Filtering, Noise Reduction, Sensor Signal Interpretation, Experimental Design

Clinical and Biomedical Data Analysis: Longitudinal Data Analysis, Repeated Measures, Clinical Outcome Analysis, Hematologic Data Analysis, Data Quality Assessment, Missing Data Review, Baseline Adjustment, Risk Estimation

Statistics: Descriptive Statistics, Median and Interquartile Range, Confidence Intervals, Wilson Score Interval, Risk Ratio, Risk Difference, Fisher's Exact Test, Categorical Data Analysis

Machine Learning: Scikit-learn, XGBoost, SHAP, Feature Engineering, Model Optimization, Dimensionality Reduction (PCA, Kernel PCA, LDA)

Data Science and Visualization: Python, Pandas, NumPy, Matplotlib, Excel Data Processing, Data Cleaning, Data Validation, Statistical Visualization, Reproducible Analysis

Engineering and Design: SolidWorks, CAD Modeling, Mechanical Design, Rapid Prototyping

EXPERIENCE

Robotics Engineer (Part-Time)

Mar 2026 – Present

University of Alberta — Edmonton, AB

- Lead ongoing development of a fleet of Raspberry Pi based robotic rehabilitation platforms, scaling deployment from 5 to a planned 12 units across the laboratory.
- Supervise two co-op mechatronics students within a 10-member lab team, reviewing code, guiding hardware debugging, and coordinating task delegation under two direct supervisors.
- Designed and shipped a secure over-the-air update pipeline for the robot fleet. A tag-triggered GitHub Actions workflow builds a validated package, computes its SHA256 checksum, generates a production manifest, and publishes a draft release that serves as the single human approval gate.
- Built the robot-side update client in Python (standard library only) with per-robot read-only fine-grained access tokens, semantic version comparison, .part downloads with SHA256 verification, structure validation, and atomic promotion. Enforced operator confirmation before any install or reboot, and validated the full flow with a real remote install and reboot on hardware.
- Applied least-privilege credential design with single-repository read-only tokens, independent per-robot revocation, and a secret-leak regression scanner covering all logs and status files.
- Wrote mocked-network test suites covering every failure branch (authentication rejection, network drop mid-download, checksum mismatch, malformed manifest), which caught bugs before hardware testing, including a network error mislabeled as a disk error.
- Reduced software-update preflight from about 2 minutes to 20 seconds (6x) and install from about 4 minutes to 30 seconds (8x). Profiled the updater with cProfile, found the same files being re-parsed 8 to 41 times per run, and added a content-hash cache, with identical validation results verified on valid and deliberately broken code.
- Diagnosed and fixed unstable enterprise Wi-Fi (WPA2-Enterprise/eduroam) on NetworkManager by tracing stale profiles and credential-storage settings through journalctl and nmcli, then added persistent auto-reconnect and a Forget Network control.

- Fixed real-time backend bugs that were discarding valid trial attempts (event ordering, dropped pending input, index errors) and a crash-report bug that reported the wrong robot ID.
- Redesigned the touchscreen Software Update workflow from 9 buttons and 13 information rows into a guided, one-button-per-step flow. The state machine reuses the existing safety gates, so the redesign can delay a button but never enable an install early.
- Built touch-optimized Tkinter components, including a scrollable RFID picker with tap-versus-drag detection, an on-screen QWERTY keyboard, a validated configuration editor with atomic writes, and live limit countdown timers, within a roughly 19,000-line application.
- Advanced the TCP-based Pi-to-Pi communication protocol for multi-robot coordination, including RFID transfer, outcome messaging, acknowledgements, and duplicate protection.
- Established semantic versioning and a documented release process, and wrote operator checklists, a safety policy, and a token-onboarding guide.
- Applied AI-assisted development tools, including GitHub Copilot and Claude, in daily implementation, debugging, and documentation, critically reviewing all AI-generated code before merging.
- Audited the logging architecture, identifying about 240 silently swallowed exceptions, in-memory-only events, and discarded backend output, and designed a structured diagnostics plan.

Robotics Engineer Intern

Nov 2025 – Feb 2026

University of Alberta — Edmonton, AB

- Developed and refined a Raspberry Pi based robotic rehabilitation platform integrating RFID identification, HX711 load cells, photo interrupter sensors, stepper motor control, pellet delivery hardware, and experimental data acquisition, taking the platform from no graphical interface to a full operator GUI.
- Built a Python/Tkinter touchscreen control interface for Raspberry Pi, integrating robot startup and stop controls, RFID management, experiment configuration, trial restrictions, live system monitoring, session statistics, Wi-Fi management, calibration workflows, and software update utilities.
- Integrated frontend and backend robot processes using controlled runtime state files and file-based inter-process communication while maintaining separation between operator controls and hardware control logic.
- Debugged hardware-software integration issues involving GPIO, SPI, UART/serial communication, RFID readers, load cells, stepper motors, sensor alignment, PCB connections, wiring, and differences across Raspberry Pi 4/5 hardware.
- Refined robot calibration workflows for motor homing, pedestal positioning, pellet detection sensors, and load cell zeroing, including operator-guided calibration steps, safety checks, and failure-specific diagnostics.
- Implemented and validated Standalone and Shuttle operating modes, with Shuttle Mode modifying post-outcome pellet presentation behavior while preserving the existing RFID, sensing, outcome analysis, and data logging pipeline.
- Developed persistent experimental data workflows for RFID-based trials, including daily data files, full historical backups, outcome tracking, session statistics, and configurable per-RFID and per-trial restrictions.
- Designed an on-device Wi-Fi management system using NetworkManager and nmcli, supporting touchscreen network discovery, secured network connection, connectivity monitoring, and WPA/WPA2 Enterprise Wi-Fi workflows.
- Engineered a guarded software deployment and update workflow using versioned packages, preflight validation, staging, protected backups, controlled installation, post-install validation, and rollback support.
- Developed the initial TCP-based Master/Slave communication components for a dual-robot architecture, including RFID transfer, pellet presentation commands, outcome messaging, and communication status monitoring.
- Improved deployment and operational reliability through GUI autostart configuration, runtime logging, error reporting, network diagnostics, and systematic laboratory testing.

PROJECTS

Clinical Data Analysis — Breast Cancer Supportive Oncology Dataset

Apr 2026 – Jun 2026

- Analyzed a longitudinal dataset of repeated hematologic measurements and chemotherapy-related adverse events across 34 breast cancer patients receiving melatonin or placebo.
- Performed systematic Excel data validation, cleaning, unit normalization, missing-data review, and consistency checks prior to statistical analysis.
- Identified and corrected inconsistent white blood cell count scaling while preserving original source values and maintaining an auditable data-cleaning workflow.
- Analyzed longitudinal trends in white blood cell count, hemoglobin, and platelet count across four treatment measurements using patient-level and group-level summaries.
- Evaluated chemotherapy-related outcomes including neutropenia incidence, chemotherapy delays, G-CSF utilization, and hospitalization associated with neutropenia and fever or infection.
- Calculated descriptive statistics, proportions, risk estimates, 95% confidence intervals, and Fisher's exact tests for low-frequency clinical outcomes.
- Built reproducible, audit-ready Excel workbooks and publication-ready visualizations, including longitudinal trend plots, baseline-adjusted change plots, risk plots, and 100% stacked distribution charts.

Standardized Biomechanical Testing Platform for Distal Biceps Fixation

Sep 2025 – Dec 2025

Master's capstone project — University of Alberta

- Designed and built a biomechanical testing platform for controlled mechanical data collection and performance analysis of distal biceps tendon fixation methods.

- Developed 3D-printed anatomical bone models and custom fixtures integrated with a BOSE ElectroForce mechanical testing system.
- Implemented cyclic loading and pull-to-failure protocols based on biomechanical literature to evaluate unicortical and bicortical fixation performance.
- Engineered reproducible testing workflows to reduce experimental variability and improve consistency of results, enabling quantitative comparison between fixation methods.
- Presented findings at the University of Alberta engineering research symposium (poster: "Testing of Repair Techniques for Distal Biceps Tendon Tears").

PredictIQ — Football Match Outcome Prediction (XGBoost)

Sep 2025 – Dec 2025

Team project (2 members) — github.com/shervinafzali/PredictIQ

- Co-developed an end-to-end machine learning pipeline including data extraction, feature engineering, model training, and evaluation.
- Designed time-aware validation strategies to prevent data leakage and simulate real-world prediction conditions.
- Applied and fine-tuned multiple ML models, optimizing an XGBoost model that achieved a macro F1 score of 0.357, outperforming baseline models.
- Applied SHAP to interpret model behavior and communicate feature importance.

Breast Cancer Subtype Classification from Microarray Data

Jan 2025 – Apr 2025

Graduate course project

- Built classification models on a high-dimensional microarray dataset (GSE45827) of 151 tumor biopsies across 54,676 genes and six subtype classes.
- Applied dimensionality reduction techniques, including PCA, Kernel PCA, and LDA, alongside correlation and Random Forest-based feature importance analysis to address multicollinearity.
- Evaluated multiple models, including SVM, Random Forest, KNN, Logistic Regression, and Gaussian Naive Bayes, achieving up to 96% accuracy and 94% macro F1 score across 5-run cross-validation.
- Performed outlier detection using Local Outlier Factor and an overfitting analysis comparing training and test performance to improve data quality and model reliability.

Android Mobile Application Development

Personal and Freelance Projects

Java

- Built Android mobile applications in Java, designing user interfaces and implementing application logic across personal and freelance projects.

Adaptive Circadian Lighting System for Neonatal Incubators

May 2022 – Jun 2022

Undergraduate capstone project — embedded firmware in C++ on Arduino

- Designed and implemented an embedded biomedical system for circadian-aligned lighting control, building the first working prototype.
- Developed microcontroller-based control logic and integrated electronic components for dynamic light regulation.
- Built and validated a complete hardware prototype through testing, debugging, and performance checks.

EDUCATION

Master of Science, Biomedical Engineering

Jan 2024 – Dec 2025

University of Alberta — Edmonton, AB

- Capstone: "Development of a Standardized Biomechanical Testing Setup for Unicortical and Bicortical Distal Biceps Cortical Button Fixation."
- Relevant coursework: Imaging Methods in Medicine, Fundamentals of Biomechanical Modelling, Machine Learning in Biomedical Applications, Biomedical Innovation, Entrepreneurship and Career, Biomedical Device Design and Technology Development, Applied Machine Learning and AI.

Bachelor of Science, Biomedical Engineering

Sep 2017 – Jun 2022

Islamic Azad University — Tehran, Iran

- Capstone: designed and built an embedded biomedical device for neonatal circadian lighting control.

CERTIFICATIONS

- Linux and SQL — Google, Issued Mar 2026 (Credential ID 2MZC731Z3D7V)
- Google Data Analytics Specialization — Google, Issued Apr 2025 (Credential ID JHLJ6YOKJQ3H)
- Master SolidWorks: Beginner to Advanced — Udemy, Issued Mar 2026

ADDITIONAL EXPERIENCE AND LEADERSHIP

- Recreation Facility Operator, University of Alberta (Sep 2024 – Present): manages operations in a high-traffic environment, ensuring safety, reliability, and real-time issue resolution while communicating clearly with staff and users.
- DRA Research Supervisor, University of Alberta: supervised undergraduate research activities, supported experimental setup development, and guided system-level problem solving.
- Event Coordination Volunteer, Engineering Graduate Research Symposium, and Event Volunteer, BioAlberta Health and Life Sciences Showcase 2025.